Tutorial 02 – Computer Arithmetic

1. **Perform the following additions between binary numbers.**
   a. 1101010 + 1100
   b. 111000 + 1010
   c. 10011011 + 110
   d. 1110000111 + 1010101
   e. 10110011 + 101010 + 11001
   f. 10101111 + 1010 + 101

2. **Perform the following subtraction between binary numbers.**
   a. 10110 - 101
   b. 111111 – 1011
   c. 10101011 - 01010
   d. 001101101 - 0011
   e. 1010100 - 10
   f. 11010101000 – 0011011

3. **Perform the following multiplication between binary numbers.**
   a. $100_2 * 010_2$
   b. $111_2 * 010_2$
   c. $110_2 * 011_2$
   d. $101_2 * 011_2$
   e. $110_2 * 101_2$
   f. $101_2 * 101_2$

4. **Find the 1's Complement and 2's Complement of the following binary numbers.**
   a. 1011
   b. 0110
   c. 101010
   d. 010010
   e. 1000 1111
   f. 1001 0001 1100
   g. 1010 0110 1000

Note: You can compare your answer by converting the values to decimal